

Review

Utility of severity scoring systems for mangled upper limb salvage: A systematic review and meta-analysis

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ABSTRACT

Background: The purpose of scoring systems is to aid in the decision-making process of whether to salvage or amputate mangled extremities, but their efficacy for upper limbs is uncertain. In this study, we examined the predictive potential of scoring systems for upper limb salvage.

Methods: Two investigators undertook a systematic search of 3 leading databases for English or Japanese literature from 1985, when the first scoring system to evaluate mangled extremities was proposed, until January 2022. To be eligible, studies must have had upper extremity limb salvage or amputation as an outcome, and identified the scoring system used and treatment outcome in individual cases.

Results: Ten studies ($N = 338$) of the Mangled Extremity Scoring System (MESS) were ultimately included in the meta-analysis. The pooled sensitivity and specificity were 0.95 (95 % CI = 0.69–0.99) and 0.81 (0.65–0.91), respectively. The area under the hierarchical summary receiver operating characteristic curve was 0.95 (0.93–0.97). A subgroup analysis showed lower specificity in isolated vascular injuries. Scoring systems other than MESS were ineligible for the quantitative synthesis because none were examined in an adequate number of publications.

Conclusion: The pooled sensitivity and specificity for MESS were comparable to those reported for the lower extremities. The specificity suggests that limb salvage was achieved in at least 20 % of the patients whose MESS was above the threshold beyond which amputation is indicated. Given the likelihood of upper extremity functional limitations following amputation and the drawbacks of prostheses, we conclude that current scoring systems poorly predict salvageability of a mangled upper extremity and should not be used to justify amputation.

Introduction

For the treatment of a mangled extremity involving multiple tissue components including bones, vessels, nerves, and soft tissue, limb salvage seems ideal for both patients and surgeons, but it is not always feasible. When the challenges of a mangled extremity are compounded by hemorrhagic shock or a severe trunk injury, systemic management and stabilization of a patient's general status is the top priority. In such cases, limb salvage must be postponed or abandoned if the life of the patient is endangered. In the reconstruction of mangled extremities with amputations or widespread skin defects, patients undergo several invasive operations, and a stable general status is essential. Even if this is well controlled, widespread soft tissue defects carry a high risk of

postoperative infection, which can result in significant functional impairment. Hence, aggressive limb salvage for the mangled extremity is not always the optimal answer. To prioritize stabilizing a patient's general status or control of hemorrhage from a mangled limb, primary amputation is considered.

Surgeons and ER doctors face the challenging decision regarding whether to perform limb salvage or primary amputation. It usually needs to be made quickly as a delay hinders the initial intervention which carries a risk of limb necrosis, infection, and fatal complications. To assist in this critically important decision-making process, scoring systems have been developed to evaluate the patient's condition, although almost all were designed specifically for the lower extremity. They include the Mangled Extremity Severity Score (MESS) [1],

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Mangled Extremity Syndrome Index (MESI) [2], and Hannover Fracture Scale [3]. The sensitivity and specificity of lower extremity scoring is reported to be 70 to 80 % [4].

The MESS was initially developed to assess mangled lower extremities, and can predict the likelihood of a lower extremity amputation. Slauterbeck et al. and Durham et al. suggested its applicability for the upper extremity as well, reporting good sensitivity and specificity [5,6]. But significant concerns have been raised in recent years about whether or not any scoring system is able to adequately predict limb amputation and salvage at all [7]. This comes amidst a trend in recent years of surgeons undertaking challenging limb salvage in the upper extremity possibly on account of factors such as the poor results of prostheses acceptance and use following primary amputation [8]. Those systems, originally created for lower extremities but adopted for use in upper extremities, of which MESS is the leading example, increasingly appear to be seen as inadequate and are being abandoned, but this claim needs verification.

We undertook this study to eliminate doubt regarding the ability of currently available scoring systems and their threshold scores to ascertain the likelihood of successful limb salvage for the mangled upper extremity. We hypothesized that existing scoring systems are not appropriate for predicting limb salvage in the upper extremity. Given that traumas causing severely mangled extremities are rare, the usefulness of scoring systems has been evaluated based on retrospective case studies in single centers only and studies including a larger patient pool have not been performed. Therefore, we tested our hypothesis by conducting a meta-analysis to examine the predictability of limb salvage in the upper extremities using currently available scoring systems.

Methods

Protocol and registration

A systematic review of studies on severity scoring systems used for mangled upper extremities was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement and the PRISMA extension for Diagnostic Test Accuracy (DTA) [9]. The study protocol was registered in PROSPERO (number CRD42021259418), which is an international database of systematic reviews, before the literature search was conducted.

Search strategy

A systematic literature search of articles published in either English or Japanese was conducted of the PubMed/MEDLINE, EMBASE, and Ichushi-Web (Japan Medical Abstracts Society) electronic databases of articles from 1985—when MESI [2], the first scoring system to evaluate mangled extremities, was proposed—until the retrieval date of January 2022. Review papers and monographs appearing in UpToDate, a computerized knowledge management system, were eligible. To identify all relevant articles, terms including *scoring*, *mangled extremity*, *upper extremity*, and *amputation* were used in the search formula (Supplemental file).

Selection criteria

In the process of screening titles, abstracts and full texts, we identified eligible studies that met the following inclusion criteria. (1) Studies must have been related to upper extremity trauma; those specifically targeting lower extremity or pelvic trauma and of tumors were not eligible. Most of the studies treated injuries to the upper and lower extremities within the same framework. When this occurred, the prerequisite for inclusion was that upper extremity cases could be precisely extracted and independently analyzed. (2) It could be confirmed that the definitive treatment available in accordance with best practice was performed. (3) The scoring threshold must have been constant and

patient outcomes above and below the threshold must have been tracked. (4) The data was specific to treatment for adults. Studies limited to pediatric cases or those that combined data irrespective of age from which adult data could not be extracted and independently analyzed were not included. (5) Studies need to have at least 10 cases to be included in the quantitative synthesis.

We excluded studies that used scoring for primary decision-making to prevent selection bias. Review articles and conference abstracts were excluded if they did not identify the scoring system and treatment outcome in individual cases. We also excluded studies focused on scoring systems unrelated to salvage decisions like the Disabilities of the Arm, Shoulder and Hand (DASH) outcome measure, unless they contained salvage decision data that could be extracted. Studies that had only one type of treatment (limb salvage or amputation) were excluded.

Quality assessment and data extraction

Two reviewers (HY and ST) independently assessed the eligibility of each study according to the selection criteria previously described. Disagreements that arose during the process were resolved by reaching a consensus after discussion. The accepted studies were assessed for risk of bias and methodology according to the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool [10]. The definition of sensitivity, the probability that patients with scores greater than or equal to the threshold underwent amputation, and specificity, the probability that patients with scores less than the threshold had limb salvage, followed that used by Schirò et al. [4]. Patients for whom an initial attempt was made to preserve the limb but ultimately underwent amputation were classified as having had an amputation. During the full text review, we also checked the types of injuries and conducted a subgroup analysis if multiple studies reported the same type of trauma.

Statistical analysis

In the meta-analysis, we used a bivariate random effects model to calculate summary estimates of the sensitivity, specificity, positive likelihood ratio (PLR), and negative likelihood ratio (NLR). A hierarchical summary receiver operating characteristics (HSROC) model was used to analyze the data. This method allows for the addition of covariates to investigate potential sources of heterogeneity, and is the standard method recommended for meta-analyses of diagnostic tests.

Heterogeneities are often found when diagnostic studies are integrated. They are commonly caused by the threshold effect. Since the scoring thresholds were constant in this study, we determined that the threshold effect was not present and formally quantified the degree of heterogeneity by calculating the heterogeneity index (using I^2 statistics) graphically displayed by forest plots. We judged that there was a statistically significant heterogeneity when the I^2 was greater than 50 %; and, if the heterogeneity was too substantial, we abandoned integrating multiple studies by means of a meta-analytic approach.

A linear regression method is commonly utilized to investigate publication bias in a meta-analysis. Deeks et al. showed that using this approach to assess publication bias in diagnostic testing studies leads to higher error rates [11]. Their procedure was to evaluate the regression of the diagnostic log odds ratio on the inverse of the square root of the effective sample size (ESS) with weighing by ESS. We used their method and determined that $p < 0.10$ indicated a significant asymmetry.

Statistical analyses were conducted with STATA 17.0 (Stata Corp., College Station, TX, USA), specifically employing the user-generated "midas" and "metandi" module [12,13]. RevMan 5.4 (The Cochrane Collaboration) was used to further evaluate bias.

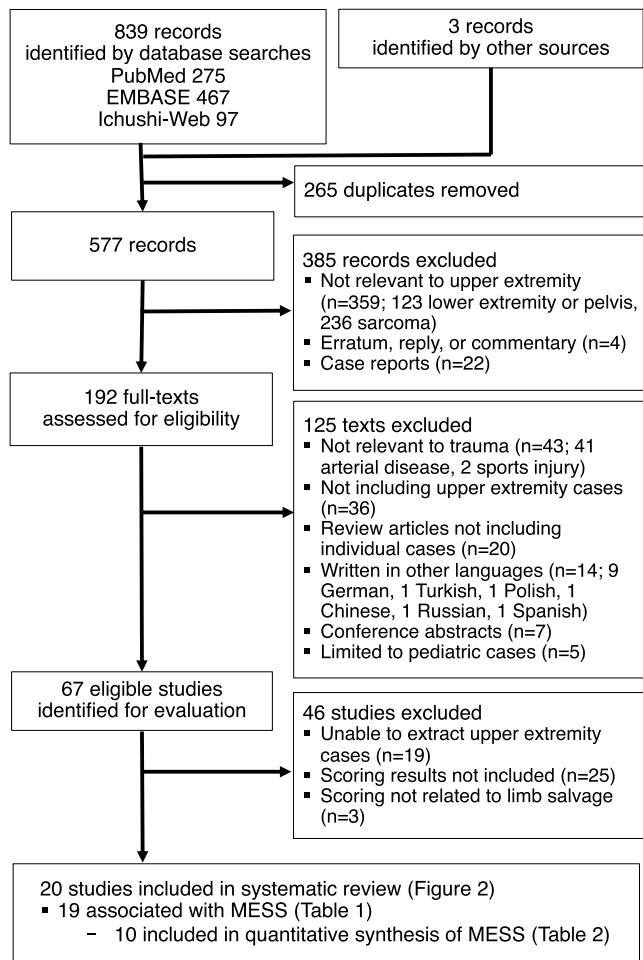


Fig. 1. Study selection process.

Results

Search results

A total of 577 records were retrieved, excluding duplicates (Fig. 1). After removing studies on sarcoma or lower extremity trauma and case reports, 192 references were eligible for full-text review. During it, studies on vascular disease and review articles from which individual cases could not be extracted were excluded, leaving 67 articles for evaluation. Upper and lower extremity data was often intermixed. Nineteen studies in which upper extremity cases could not be isolated for analysis were excluded. Additionally, 25 studies that did not include scoring and 3 studies that included scoring not related to limb salvage were also excluded. This process yielded 20 publications for the systematic review.

We were able to extract upper extremity cases from 19 studies on the

Table 2
Diagnostic performance of MESS for primary amputation (10 studies).

Author, Year	Sensitivity	Specificity	PLR	NLR
Slauterbeck [5], 1994	1.00 (0.66–1.00)	1.00 (0.89–1.00)	N/A	0.00 (N/A)
Durham [6], 1996	1.00 (0.77–1.00)	0.89 (0.52–1.00)	9.00 (7.15–10.85)	0.00 (N/A)
Nakamura [23], 1998	1.00 (0.15–1.00)	0.36 (0.11–0.69)	1.57 (1.12–2.02)	0.00 (N/A)
Huang [24], 2007	1.00 (0.03–1.00)	0.73 (0.45–0.92)	3.75 (2.91–4.59)	0.00 (N/A)
Simmons [25], 2008	0.50 (0.07–0.93)	0.92 (0.78–0.98)	6.17 (5.70–6.63)	0.54 (0.20–1.46)
Prichayudh [26], 2009	1.00 (0.59–1.00)	0.73 (0.58–0.85)	3.75 (3.27–4.23)	0.00 (N/A)
Bakker [27], 2013	1.00 (0.03–1.00)	0.89 (0.67–0.99)	9.50 (8.19–10.81)	0.00 (N/A)
Ege [28], 2015	0.80 (0.28–0.99)	0.84 (0.64–0.95)	5.00 (4.22–5.78)	0.24 (0.04–1.39)
Fochtman [29], 2016	1.00 (0.63–1.00)	0.80 (0.66–0.91)	5.11 (4.53–5.70)	0.00 (N/A)
Savetsky [30], 2019	0.71 (0.29–0.96)	0.36 (0.21–0.53)	1.11 (0.66–1.88)	0.80 (0.23–2.76)

PLR, positive likelihood ratio; NLR, negative likelihood ratio; N/A, not available. Numbers in parentheses indicate the 95 % confidence interval calculated by the Clopper–Pearson method.

Table 1
Characteristics of studies eligible for systematic review of MESS (19 Studies).

Author, Year	Patients	Average age	MESS≥7 Amputation	Salvage	MESS<7 Amputation	Salvage	Quantitative synthesis
Slauterbeck [5], 1994	43	N/A	9	0	0	34	Yes
Durham [6], 1996	23	N/A	14	1	0	8	Yes
Nakamura [23], 1998	13	34.4	2	7	0	4	Yes
Togawa [14], 2005	3	43.3	1	2	N/A	N/A	
Chambers [19], 2006	9*	N/A	6	2	0	1	
Sané [20], 2007	9	39.5	2	0	0	7	
Huang [24], 2007	16	38.8	1	4	0	11	Yes
Simmons [25], 2008	41	N/A	2	3	2	34	Yes
Prichayudh [26], 2009	52	28.7	7	12	0	33	Yes
Korompilias [15], 2009	3	19.0	2	N/A	1	N/A	
Bakker [27], 2013	20	28.8†	1	2	0	17	Yes
Ege [28], 2015	30	24.6†	4	4	1	21	Yes
Fochtman [29], 2016	54	38.0	8	9	0	37	Yes
Kumar [16], 2017	10	35.8	3	7	N/A	N/A	
Karakus [22], 2017	288	N/A	30	0	258	0	
Kansal [21], 2019	6	41.5	3	0	0	3	
Savetsky [30], 2019	46	N/A	5	25	2	14	Yes
Hohenberger [17], 2020	42	40.3†	N/A	7	2	33	
Hohenberger [18], 2020	14	34.4†	N/A	4	N/A	10	

MESS, Mangled Extremity Severity Score; N/A, not available.

* Number of cases in which prognosis could be identified from the text.

† Mean age of all patients included in the study.

	Risk of Bias				Applicability Concerns		
	Patient selection	Index text	Reference standard	Flow and timing	Patient selection	Index text	Reference standard
Slauterbeck [5], 1994	😊	😊	😊	😊	😊	😊	😊
Durham [6], 1996	😊	😊	😬	😊	😊	😊	😬
Dzepina [29], 1997	😊	😊	😊	😊	😊	😊	😊
Nakamura [23], 1998	😊	😊	😊	😊	😊	😊	😊
Togawa [14], 2005	😊	😊	😊	😊	😊	😊	😊
Chambers [19], 2006	😞	😊	😬	😊	😬	😊	😬
Sane [20], 2007	😊	😊	😊	😊	😊	😊	😊
Huang [24], 2007	😊	😊	😊	😊	😊	😊	😊
Simmons [25], 2008	😊	😊	😊	😊	😊	😊	😊
Prichayudh [26], 2009	😊	😊	😊	😊	😊	😊	😊
Korompilias [15], 2009	😞	😊	😊	😊	😞	😊	😊
Bakker [27], 2013	😊	😊	😬	😊	😊	😊	😬
Ege [28], 2015	😊	😊	😊	😊	😊	😊	😊
Fochtmann [29], 2016	😊	😊	😊	😊	😊	😊	😊
Kumar [16], 2017	😞	😊	😊	😊	😞	😊	😊
Karakus [22], 2017	😊	😞	😞	😊	😊	😞	😞
Kansal [21], 2019	😊	😞	😞	😊	😊	😞	😞
Savetsky [30], 2019	😊	😊	😊	😊	😊	😊	😊
Hohenberger [17], 2020	😞	😊	😊	😞	😞	😊	😊
Hohenberger [18], 2020	😞	😊	😬	😞	😞	😊	😬

😊 = Low, 😬 = Unknown, 😞 = High

Fig. 2. Methodological quality according to the Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool. The table shows the risk of bias and applicability concerns for each study.

MESS, the scoring system used most often according to the literature (Table 1). There were too few studies using scoring systems other than MESS to attain an adequate sample size, which eliminated the possibility of conducting a meta-analysis on them. Not all the MESS-related literature was suitable to undergo a quantitative synthesis in our meta-analysis and some studies were excluded. Specifically, we excluded 5 studies for which it was not possible to generate a 2 × 2 contingency table based on the reported outcomes; among them, studies that only targeted patients with a MESS of 7 or higher and studies that examined either salvage or amputation but not both [14–18]. We excluded one study in which the prognosis of all patients could not be tracked [19] and two studies reporting outcomes of fewer than 10 patients that were of insufficient power to be used in the meta-analysis [20,21]. Finally, a study with more than 288 cases by Karakus et al. had to be excluded because MESS was used for primary treatment decision-making [22]. Ultimately, ten studies could be included in a meta-analysis on the accuracy of the MESS [6,23–30]. Their characteristics are shown in Table 2.

We carefully assessed the quality of all included studies using QUADAS-2. Concerns were exposed in the primary diagnostic accuracy

of certain studies, notably in the third domain (Reference Standard) regarding the risk of bias and concerns about applicability, as represented in a cumulative bar graph (Figs. 2 and 3). For example, there was a potential risk of reporting bias in a study in which patient outcome was assessed by a telephone survey. Also, studies with a narrow focus such as one limited to vascular injury might have applicability concerns in patient selection. Such studies that might include potential bias or concerns were carefully reviewed by two examiners and ultimately not excluded.

Diagnostic accuracy of MESS for predicting upper limb salvage and amputation

Ten studies with 338 patients in total were used in the meta-analysis to validate the diagnostic accuracy of MESS for limb salvage of mangled upper extremities. The summarized sensitivity and specificity were 0.95 (95 % CI= 0.69–0.99) and 0.81 (95 % CI= 0.65–0.91), respectively (Fig. 4). The area under the HSROC was 0.95 (95 % CI= 0.93–0.97; Fig. 5). The positive and negative likelihood ratios were 5.1 (2.4–10.5) and 0.06 (0.65–0.91), respectively. The heterogeneity was not

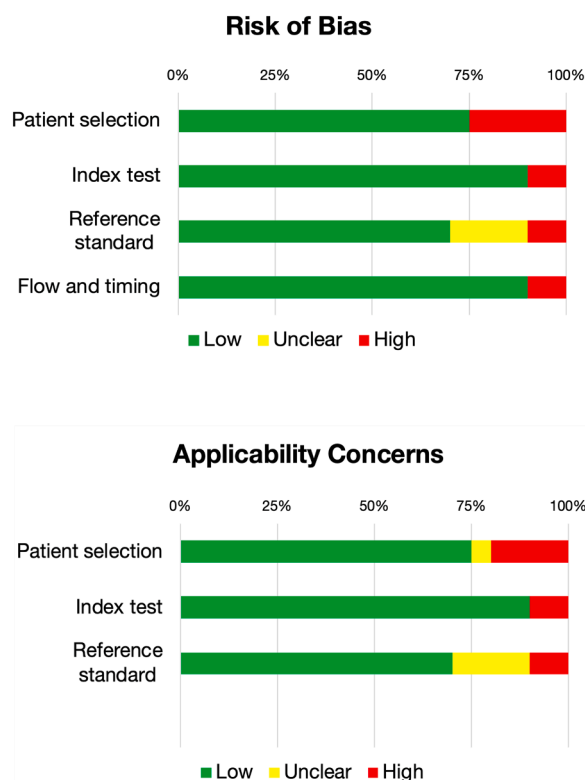


Fig. 3. Summary of methodological quality of studies. Graphs show percentages of the risk of bias and applicability concerns in the total included studies.

significant ($I_2 = 10\%$). We had planned to examine the sensitivity and specificity by changing the threshold, but the limited number of articles describing MESS and outcomes for individual patients prevented us from conducting a power analysis.

Subgroup analysis of MESS

Of the 18 articles from which we could retrieve upper extremity trauma cases, five were specific to isolated vascular injuries [15,18,20,25,26] and two were specific to injuries around the elbow [20,24]. A qualitative evaluation was conducted in the subgroup analysis; but, on account of missing values, it was impossible to perform a quantitative integration. The reported sensitivities and specificities for injuries limited to vessels were 67–100 % and 33–92 %, respectively, with specificity trending downward. False positives were higher in cases of isolated vascular injuries, suggesting that even high MESS values may increase the likelihood of salvage. There were no characteristic results for injuries around the elbow.

Publication bias of MESS

According to Deeks' test, there was no publication bias. In the forest plot created by graphing the logarithm of the diagnostic odds ratio and the inverse of the root of the effective sample size, no asymmetry was detected (Fig. 6). We presume that publication bias was not present in this meta-analysis.

Diagnostic accuracy of other scoring systems

In addition to the MESS, we searched for literature on the MESI, Hannover Fracture Scale, and Mangled Upper Extremity Score (MUES) to extract mangled upper extremity scoring. There were five publications from which upper extremity cases could be acquired, four for the MESI [6,15,16,31] and one for the MUES [30] (Tables 3 and 4). There

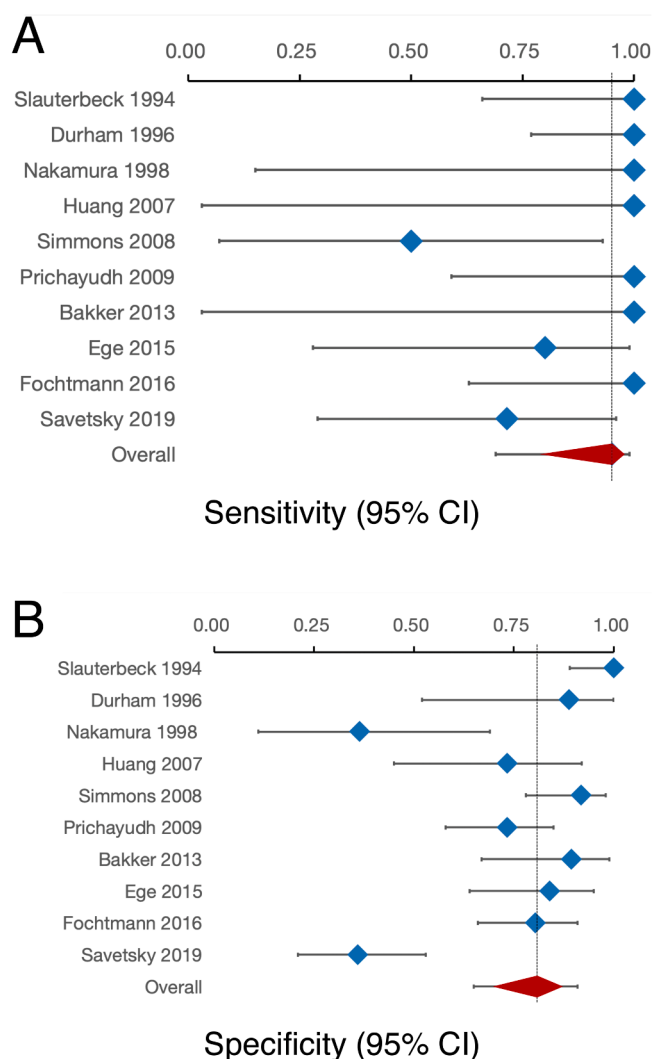


Fig. 4. Forest plots showing utility of MESS for predicting amputation and limb salvage. A, sensitivity; B, specificity. Horizontal lines represent 95 % confidence intervals.

were no studies of the Hannover Fracture Scale from which upper extremity cases could be extracted. We were unable to obtain the four or more studies required to use a quantitative approach and conduct a meta-analysis of any scoring system other than the MESS.

Studies including MESI reported a sensitivity between 86 and 100 % and a specificity between 83 and 100 %. Upon introducing the MUES scoring system in 2019, Savetsky et al. [30] reported that it had a sensitivity of 43 % and specificity of 82 %; however, its diagnostic accuracy had yet to be confirmed by additional studies at the time of our literature retrieval.

Limitations

This study has several limitations. First, we conducted the systematic review using retrospective studies that may have included reporting bias. Second, there were many studies with combined upper and lower extremity injuries from which upper extremity injuries could not be extracted that consequently had to be excluded. Third, the MESS threshold could not be controlled because studies frequently did not provide data for individual cases. If it had been possible to change the threshold, it might have served as a tool to assist in decision-making for primary amputation by increasing specificity at the expense of sensitivity. Fourth, there is a possibility of heterogeneity effects, i.e.

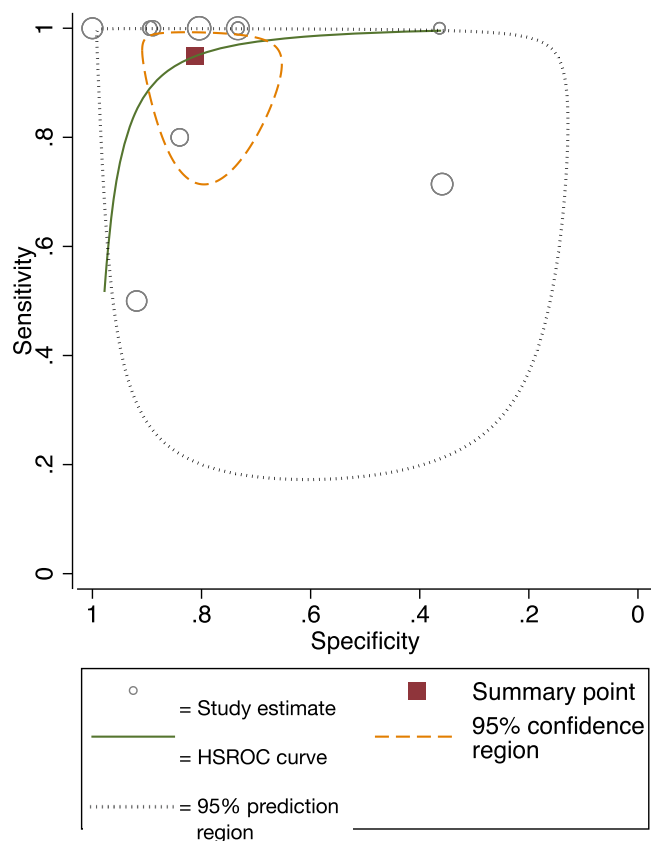


Fig. 5. MESS hierarchical summary receiver operating characteristic (HSROC) curves for predicting amputation. The circle size indicates the weight of individual studies. A square on the curve represents summary sensitivity and specificity. The area enclosed by the dashed line is the 95 % confidence interval of the summary estimate. It represents the range within which the summary sensitivity and specificity are estimated to lie within the 95 % confidence interval. The area separated by the dots is the 95 % prediction region. It indicates the range within which the data points of future studies, should they be performed are predicted to appear with 95 % accuracy.

differences in the level of treatment between institutions.

Discussion

The decision whether to perform primary amputation or limb salvage has a significant impact on the patient's functional ability, as well as their social and psychological health, all impacting their quality of life. Surgeons have a pivotal, anguishing decision to make in a very short time. Limb salvage is an ideal for both surgeons and patients, but without proper planning it can result in serious, even fatal complications and place physical and economic burdens on patients. Scoring systems have been developed to aid responsible decision-making but how useful those currently available for the upper extremity are is unsettled. Among them only MESS had a sufficient number of studies associated with it to investigate meta-analytically. Studies on MESS itself posed their own challenges. During the analysis, we found that the results for upper and lower extremities were combined in many studies, limiting the number by which to evaluate the upper extremity independently.

The meta-analysis showed that upper extremity limb salvage is highly likely when MESS is below the salvage threshold. However, with its low specificity and high false positives, MESS should not be the sole benchmark for primary amputation. We found that approximately 20 % of patients who underwent an amputation of an upper extremity had the possibility of limb salvage. This might not be surprising given the inferior specificity of the MESS reported by studies in the past decade [17,

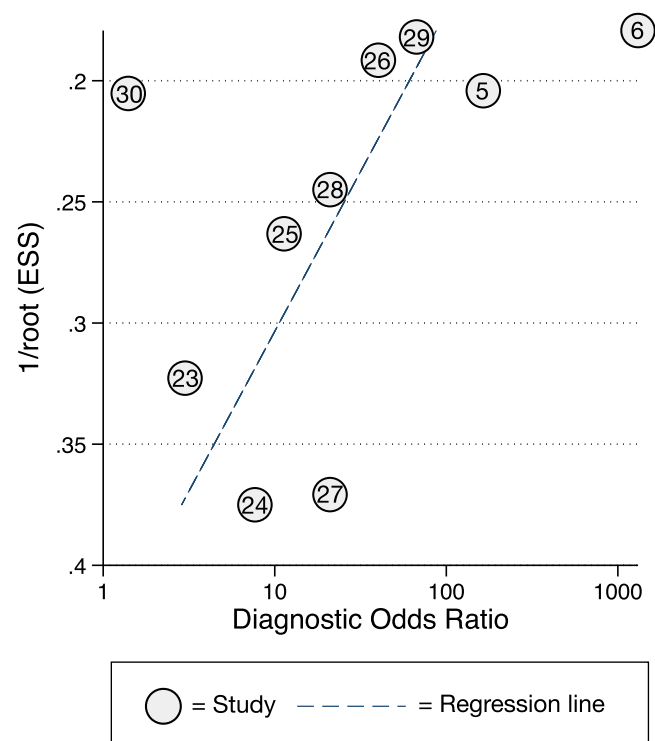


Fig. 6. Assessment of publication bias using the Deeks funnel plot asymmetry test [10]. The number corresponds to the literature number of each study. Diagnostic log odds ratios are regressed against the inverse of the square root of the effective sample size and weighted by the effective sample size, yielding a funnel plot with the regression line overlapping the center. The symmetry of the data suggests a low likelihood of publication bias. —ESS, effective sample size; $P = 0.17$.

18,29,30]. This reflects advances in damage control techniques, which affords more patients the chance to undergo reconstructive surgery aimed at limb salvage [32,33]. The accuracy of MESS in the lower limb has been reported in a large study of 556 cases and in a systematic review. Its sensitivity varied widely, ranging from 45.6 to 100 %, and its use in deciding when to amputate was controversial [4,34]. This indicates that the utility of MESS is unacceptable not only for the upper mangled extremity but also for the lower one.

As we were unable to compile a statistically viable number of studies to properly examine the value of scoring systems other than MESS with sufficient quality, we could not conclude that any of them were appropriate for the upper extremity. Considering that scoring systems currently in use for mangled upper extremities are inadequate and may even mislead ER physicians and reconstructive surgeons, it is time to critically evaluate them.

We contend that the threshold for primary amputation should be elevated to increase the potential of limb salvage, remaining cautious to avoid harming the patient. Ischemia following trauma to mangled upper extremities is rarer than in the lower extremities due to the greater density of collateral arteries [35], which decreases the risk of necrosis and reperfusion injury. We were mindful of this when our subgroup analysis showed a trend toward lower specificity in cases of vascular injury alone, with the likelihood of salvage increasing in them even when the scoring was above the threshold. This lowered the threshold for salvage of the upper limb. Moreover, because of the considerable limitations of hand prostheses, limb salvage had better patient-reported outcomes compared to amputation, even if the functional result of the reconstruction was very limited [8]. This differs from the conclusions reached by the Lower Extremity Assessment Project (LEAP) study and follow on studies, which found no difference in health status or function between limb salvage and amputation [35,36]. While secondary

Table 3
Characteristics of studies eligible for meta-analysis of scoring systems other than MESS.

Author, Year	Scoring	Patients	Average age	Scoring ≥ threshold value		Scoring < threshold value	
				Amputation	Salvage	Amputation	Salvage
Durham [6], 1996	MESI	23	N/A	12	0	2	9
Dzepina [31], 1997	MESI	18	32.1	N/A	3	N/A	15
Korompilias [15], 2009	MESI	3	19.0	3	N/A	0	N/A
Kumar [16], 2017	MESI	10	35.8	3	7	N/A	N/A
Savetsky [30], 2019	MUES	46	N/A	3	7	4	32

MESI, Mangled Extremity Syndrome Index; MUES, Mangled Upper Extremity Severity Score; N/A, not available.

Table 4
Diagnostic performance of scoring systems other than MESS for primary amputation.

Author, Year	Scoring	Sensitivity	Specificity	PLR	NLR
Durham [6], 1996	MESI	0.86 (0.57–0.98)	1.00 (0.66–1.00)	N/A	0.14 (0.04–0.52)
Dzepina [31], 1997	MESI	N/A	0.83 (0.59–0.96)	N/A	N/A
Korompilias [15], 2009	MESI	1.00 (0.29–1.00)	N/A	N/A	N/A
Kumar [16], 2017	MESI	N/A	N/A	N/A	N/A
Savetsky [30], 2019	MUES	0.43 (0.10–0.82)	0.82 (0.66–0.92)	0.81 (7.08–7.08)	0.70 (0.36–1.34)

PLR, positive likelihood ratio; NLR, negative likelihood ratio; MESI, Mangled Extremity Syndrome Index; MUES, Mangled Upper Extremity Severity Score; N/A, not available.

amputation subsequent to a salvage attempt may be indicated in very limited situations such as necrosis, severe postoperative infection, or uncontrollable pain like complex regional pain syndrome (CRPS) [37], the risk should not be among the primary concerns. Even when patients lack hand function altogether following a traumatic upper extremity injury, they tend to accept the result compared to the alternative. The possibility of debilitating pain sequelae, such as CRPS, are not factored in because they cannot be predicted at the point of decision-making.

The MUES, an algorithm for determining when an upper extremity salvage attempt or amputation is indicated, has been proposed recently [30]. Skin and bone defects and crush injury mechanisms that can affect complications in upper limb reconstruction are included in the assessment. MUES showed a higher specificity than MESS in a scoring study targeting the same patient group [30]. Additional studies are needed to confirm its utility. Scoring systems for the mangled extremity usually do not correlate with postoperative function, but scoring such as the MUES that include factors directly leading to complications may be able not only to predict the probability of salvage but also to predict the patient's future limb function.

We identified the severe limitations of current scoring systems, and explained why they should not be used prognostically. However, we envision that the core tenets of the MESS, in a revised form, could have value in decision-making specifically related to damage control surgery for the upper extremity, assuming limb preservation in principle. Such a tool would preserve some of the current clinical variables such as limb ischemia time and shock. Proposed modifications would include elevating the upper age threshold indicating high-risk from 50 to 70 or even 80 years, given advances in systemic management techniques. It would also account for the fact that multiple surgeries are usually required for limb salvage and that older patients typically require longer treatment periods [38,39]. New parameters would be introduced, such as a patient-centered interaction where patients or their proxies would be informed about the significant morbidity and mortality risk associated with limb salvage as well as the likelihood of multiple revision surgeries and rehabilitation periods, including the extent to which it could be physically and emotionally demanding on them. Finally, it would more clearly incorporate the wishes of the patient regarding whether they want to proceed with limb salvage or not.

In summary, we conclude that the algorithmic scoring systems used to evaluate, predict outcomes and make decisions regarding treatment of mangled upper extremities—most decades old and not developed for the upper extremity—fail to meet acceptable standards justifying their use. Surgeons should understand that these systems are bound by

significant inherent limitations regarding their ability to predict the likelihood of success or failure of limb salvage attempts. They ought not be given undue precedence. If used in decision-making, it should be remembered that they are incapable of dictating an accurate threshold value for when a primary amputation is indicated. While an aggregation of indices might have greater clinical utility for predicting functional outcomes and suggesting whether to salvage or amputate, there is currently no gold standard for decision-making. We call for the creation of a new, validated scoring system explicitly for the upper extremity with a predictive value demonstrated to have a high specificity and sensitivity to reduce the burden on the medical team charged with this decision.

CRedit authorship contribution statement

Hidemasa Yoneda: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Shinsuke Takeda:** Data curation. **Masaomi Saeki:** Data curation. **Katsuyuki Iwatsuki:** Data curation. **Michiro Yamamoto:** Supervision, Project administration. **Masahiro Tatebe:** Supervision. **Hitoshi Hirata:** Supervision.

Declaration of competing interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.injury.2024.111447](https://doi.org/10.1016/j.injury.2024.111447).

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